Topic: The Impact of Conditional Cash Transfers (SURE-P MCH Program) on Maternal Autonomy in Nigeria
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Introduction

Despite the progress and expansion in access to health services in general in Nigeria, the country still records a quite high maternal mortality rate. Beyond medical factors which have been studied widely, women's ability to make independent decisions about their health plays a critical role in their health-seeking behaviors. Conditional Cash Transfers (CCT) programs are used in LMICs (Low- and Middle-Income Countries) to encourage the use of healthcare services, however not much research has been done to understand if such incentives shift underlying Gender norms and women's autonomy. A range of less commonly reviewed outcomes in reviews include household expenditure, poverty, empowerment, political participation and domestic violence (Francesca Bastagli et al, 2016).

In sub-Saharan Africa, growing evidence also suggests that cash transfers can influence intra-household power dynamics, reducing gender-based violence and increasing women's control over resources (Hidrobo Peterman & Lori Heise, 2016) (Haushofer & Shapiro, 2016). This paper seeks to examine the impact of the Subsidy Reinvestment & Empowerment Program – Maternal and Child Health (SURE-P MCH) CCT scheme which was launched in 2012 in Nigeria improved women's decision-making autonomy within households.

Using the DHS data from 2008 (pre-program) and 2013 & 2018 (post-program), employing a DiD (difference-in-differences) design comparing between states that implemented the CCT program with those that did not. The analysis provides evidence on whether targeted financial incentives can generate broader empowerment effects beyond healthcare utilization and contributes to the growing body of research on the empowerment effects of cash transfers by focusing on the African/Nigerian context and by examining both short and long-run dynamics of change following program implementation.

Background

Nigeria as a developing country still faces high rates of maternal and child mortality partly linked to the low utilization of antenatal care and health facility services. Generally, cash transfers (CCTs) are used in developing countries to incentivize health service utilization. However, the impact of these CCTs on women's autonomy which is a deeper social and behavioral outcome, is less studied. This study provides evidence from Nigeria, using a national program to evaluate whether these financial incentives shift household decision-making power.

The Subsidy Reinvestment and Empowerment Program (SURE-P) was introduced in 2012 by the Federal Government of Nigeria following partial fuel subsidy removal. The Maternal and Child Health (MCH) project, included a Conditional Cash Transfer (CCT) scheme where pregnant women received cash transfers conditional on completing a series of health visits such as registering for antenatal care (ANC), attending four ANC visits, delivering in a health facility and for completing postnatal care visits. The objective was to reduce financial barriers, incentivize facility-based delivery, and improve maternal and child health outcomes. (Oduenyi & Ordu & Okoli, 2019) The program was supported by federal resources and implemented in partnership with selected states with high maternal mortality and weaker health service uptake. The main implementation states were:

- Anambra
- Bauchi
- Kaduna
- Niger
- Ondo
- Zamfara

This selective rollout creates a natural experiment for studying causal impacts. By comparing changes in autonomy before and after the program in treatment versus control states, I assess whether the SURE-P CCT scheme influenced maternal autonomy outcomes.

Literature review

Cash transfer programs have become a central part of social-protection policy in many developing countries, designed to reduce poverty and improve health and education outcomes. Early evidence from Latin America, particularly Mexico's Oportunidades program, showed that conditional cash transfers (CCTs) can significantly increase prenatal care, school attendance, and nutritional outcomes (Barber S. & Gertler P., 2008) (Fiszbein A. & Schady N., 2009). Systematic reviews also confirm consistent positive effects of CCTs on human-capital investment and poverty reduction (Francesca Bastagli et al, 2016).

Recent work has shifted toward understanding the gendered effects of cash transfers. Evidence from Ecuador and Kenya indicates that programs targeting women can enhance their control over household resources and reduce intimate-partner violence (Hidrobo Peterman & Lori Heise, 2016) (Haushofer & Shapiro, 2016). (Peterman A. Yumkella F. & Cook S., 2017) also further demonstrate that social transfers improve women's bargaining power and reduce household conflict across sub-Saharan Africa.

In Nigeria, women's autonomy has been identified as a key determinant of maternal and child health outcomes. Studies using DHS data find that higher female education, wealth, and labor-market participation are strongly correlated with decision-making power and the likelihood of facility-based delivery (Obasohan P. E., 2019) (Ariyo et al., 2020) (Imo et al., 2022).

Evaluations of the SURE-P Maternal and Child Health program provide evidence of increased antenatal registration and institutional deliveries (Onwujekwe et al., 2020), but the effect on women's autonomy remains under-explored. This study contributes to that literature by applying a causal difference-in-differences framework to assess whether the SURE-P CCT program strengthened women's decision-making autonomy between 2008 and 2018.

Research Design

This study evaluates the causal impact of the SURE-P MCH conditional cash transfer (CCT) program, launched in 2012 in Nigeria, on women's decision-making autonomy using a difference-in-differences (DiD) approach using three survey rounds (2008-pre-program, 2013-short-run post, and 2018-long-run post).

- **Outcome (Y_{ist}):** Autonomy index for women *'i'* in state *'s'* and year *'t'* gotten from the DHS survey questions:
  I. Who decides on woman's own healthcare (v743a)
  II. Who decides on large household purchases(v743b)
  III. Who decides on visits to relatives/friends (v743d)

  {Each question is coded 1 if the woman decides alone or jointly and 0 otherwise. The average of this forms the autonomy

- **Treatment: ($Treat_s$):** whether a woman lives in a state where SURE-P was implemented (Anambra, Bauchi, Kaduna, Niger, Ondo, Zamfara).
- **Post periods ($Post_t$):** Dummy variables for the post-program years (2013 and 2018), with 2008 as the basline.
  - post2013 = 1 for survey year 2013 (immediate post-intervention),
  - post2018 = 1 for survey year 2018 (long-term effect).

- **Control variables:** The regression controls for individual-level characteristics that may influence autonomy, including age (v012), education level (v106), household wealth index (v190), and urban residence (v025). State-level (v024) and year fixed effects are included to account for time-invariant regional factors and common temporal shocks.

  All estimates are weighted using DHS sampling weights (v005) to ensure national representativeness. Standard errors are robust and clustered at the state level.

Empirical Strategy

The empirical approach follows a Difference-in-Differences (DiD) framework to identify the causal effect of the SURE-P program on women's autonomy. The model is specified as:

The DiD regression equation would be:

$$\boldsymbol{Y_{ist} = \alpha + \beta_{2013}(Treat_s \times Post_{2013}) + \beta_{2018}(Treat_s \times Post_{2018}) + \gamma X_{ist} + \delta_s + \lambda_t + \varepsilon\_ist}$$

$\boldsymbol{X_{ist}}$ are controls for woman's age, education, wealth, etc.
$\boldsymbol{\delta_s}$ are state fixed effects (to absorb time variant differences across states.
$\boldsymbol{\lambda_t}$ are survey year fixed effects
Errors are clustered at state level and DHS sampling weights are used.
The coefficients are;
$\boldsymbol{\beta_{2013}}$ = short-run effect (just after program roll-out)
$\boldsymbol{\beta_{2018}}$ = long-run effect (to check if changes persisted or grew over the years)

Identification and Validity

The central identifying assumption of the DiD framework is the parallel trends assumption that in the absence of SURE-P, women's autonomy in treated and control states would have followed similar trajectories over time. Although only one pre-program year (2008) is available, the similarity in pre-trends shown in Figure 1 provides supporting evidence. Additional robustness is achieved by including controls and fixed effects, which help adjust for observable differences across states.

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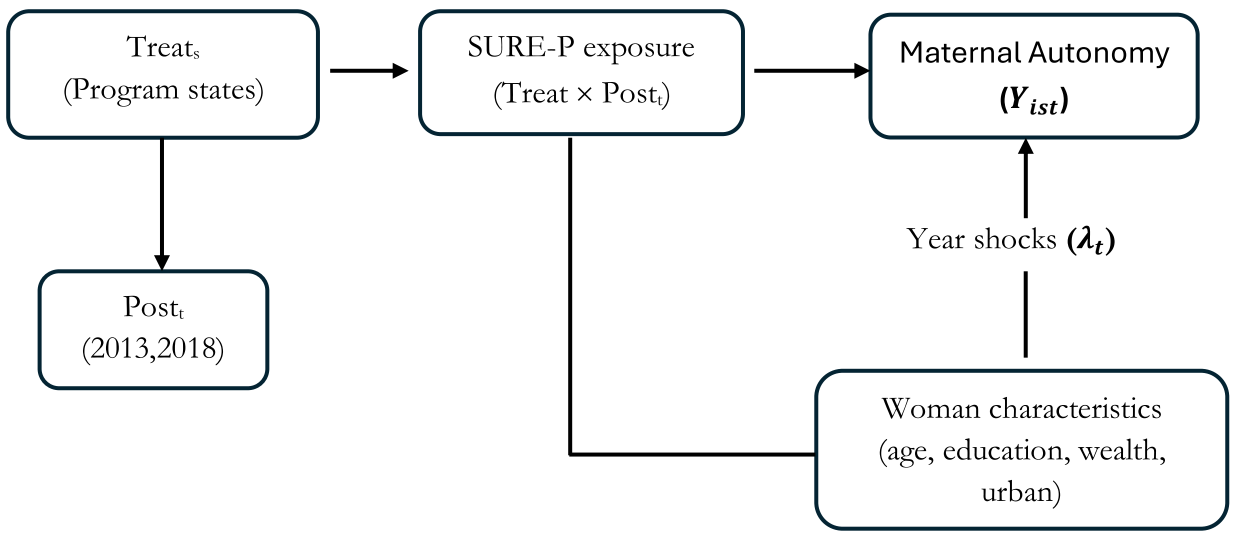

Data

The main analysis uses pooled cross-sectional data from the Nigeria DHS Individual Recode (IR) files. I estimate weighted regressions using DHS sample weights and cluster standard errors by state to account for within-state correlations. As robustness checks, I will (i) estimate models with and without controls, (ii) test sensitivity to alternative autonomy definitions, and (iii) explore heterogeneity by education and urban–rural residence to assess whether empowerment effects vary across subgroups.

This research design compares changes in women's autonomy between SURE-P and non–SURE-P states before and after program introduction. The inclusion of both 2013 and 2018 DHS rounds enables assessment of whether program effects persisted or evolved over time. The next section presents descriptive statistics and regression results examining the impact of the SURE-P CCT program on maternal autonomy.

Descriptive summary: N mean sd min max by(survey_year)

survey_year:

2008	N	Mean	SD	Min	Max
Autonomy index	23890	.45	0.43	0	1
Age	33385	28.65	9.49	15	49
Educational level	33385	1.09	1.02	0	3
Wealth Index	33385	2.92	1.42	1	5
Urban	33385	.31	0.46	0	1
2013					
Autonomy index	27222	.43	0.44	0	1
Age	38948	28.86	9.69	15	49
Educational level	38948	1.21	1.03	0	3
Wealth Index	38948	3.12	1.39	1	5
Urban	38948	.4	0.49	0	1
2018					
Autonomy index	28888	.48	0.43	0	1
Age	41821	29.16	9.71	15	49
Educational level	41821	1.26	1.04	0	3
Wealth Index	41821	3.03	1.39	1	5
Urban	41821	.41	0.49	0	1

Table 1. Descriptive Statistics of Key Variables by Survey Year

Weighted means and standard deviations of selected individual and household characteristics from the Nigeria DHS 2008–2018. The sample includes ever-married women aged 15–49.

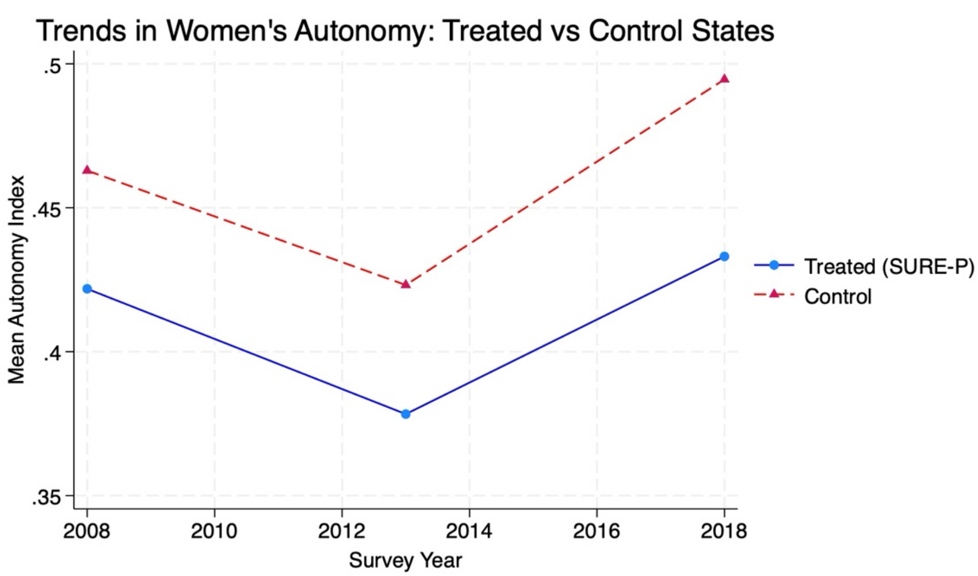

Figure 1. Trends in Women's Autonomy, Treated vs Control States

Average autonomy index by survey year for SURE-P (treated) and non-SURE-P (control) states, weighted by DHS sampling weights. Parallel pre-trends are evident before 2013.

Treated states already had slightly lower autonomy (≈0.42) than control states (≈0.46). This baseline gap is small but consistent, supporting the parallel-trends assumption (i.e. both groups had similar trajectories before the program).

Autonomy dipped for both groups in 2013, which may reflect external factors like economic conditions, survey timing, or general social factors. Importantly, both lines declined in roughly parallel fashion, again suggesting that the groups were following similar paths.

Autonomy increased again for both treated and control states. However, control states improved more sharply (≈0.49 vs. 0.43), suggesting that:

- Women's decision-making power improved nationally (not only in treated areas) due to other factors.
- The SURE-P program's direct effect on autonomy may have been limited.

The descriptive results and trend patterns presented in the graph above suggests that women's autonomy followed similar trajectories across treated and control states prior to the SURE-P rollout, consistent with the parallel-trends assumption required for Difference-in-Differences estimation. However, the modest post-2013 divergence in autonomy implies that any program-specific effects may have been limited or gradual. To formally test this, the next section estimates a series of DiD regression models that compare changes in women's autonomy between SURE-P and non-SURE-P states before and after the program, controlling for individual characteristics and state- and year-level fixed effects.

Impact of CCT (SURE-P) on Women's Autonomy

	(1)	**(2)**
VARIABLES	**Baseline Model**	**Controlled Model**
1 = SURE-P Treated State = 1	-0.0411***	0.0437***
	[0.0076]	[0.0071]
Post-2013 Period = 1	-0.0397***	-0.0174***
	[0.0049]	[0.0040]
0b.treat#0b.post2013	0.0000	0.0000
	[0.0000]	[0.0000]
0b.treat#1o.post2013	0.0000	0.0000
	[0.0000]	[0.0000]
1o.treat#0b.post2013	0.0000	0.0000
	[0.0000]	[0.0000]
1.treat#1.post2013	-0.0038	-0.0057
	[0.0109]	[0.0100]
Post-2018 Period = 1	0.0317***	0.0291***
	[0.0048]	[0.0040]
0b.treat#0b.post2018	0.0000	0.0000
	[0.0000]	[0.0000]
0b.treat#1o.post2018	0.0000	0.0000
	[0.0000]	[0.0000]
1o.treat#0b.post2018	0.0000	0.0000

	[0.0000]	[0.0000]
1.treat#1.post2018	-0.0204*	-0.0067
	[0.0107]	[0.0095]
Respondent's Age		0.0051***
		[0.0002]
Education Level		0.0671***
		[0.0023]
Wealth Index		0.0184***
		[0.0017]
Urban Residence (1=urban)		-0.0082**
		[0.0040]
Region = 2, north east		-0.1327***
		[0.0056]
Region = 3, north west		-0.2148***
		[0.0051]
Region = 4, south east		0.1168***
		[0.0067]
Region = 5, south south		0.1359***
		[0.0063]
Region = 6, south west		0.1578***
		[0.0060]
survey_year = 2013, omitted		-
survey_year = 2018, omitted		-
Constant	0.4629***	0.2033***
	[0.0034]	[0.0081]
Observations	80000	80,000
Observations	80,000	80000
Adjusted R-squared	0.0065	0.2753
F-statistic	1942	1942

Robust standard errors in brackets
*** p<0.01, ** p<0.05, * p<0.1
Robust standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01

Table 2. Impact of the SURE-P Conditional Cash Transfer (CCT) Program on Women's Autonomy: Baseline and Controlled Models

The dependent variable is the women's autonomy index, constructed from decision-making indicators in the DHS (e.g., decisions on health care, large household purchases, and visits to relatives). Column (1) presents baseline difference-in-differences estimates without additional controls. Column (2) includes controls for respondent age, education, household wealth, and urban residence, as well as regional fixed effects. Robust standard errors are in parentheses.

Results and Discussion

Table 2 and Figure 1 present the descriptive and regression results on women's decision-making autonomy across the 2008, 2013, and 2018 DHS surveys, distinguishing between states that implemented the SURE-P maternal health conditional cash transfer (CCT) program and those that did not. Overall, autonomy levels varied modestly across years. In 2008, the mean autonomy index was 0.42 in treated states compared to 0.46 in control states. By 2013, autonomy declined slightly in both groups, before rising again in 2018. These trends suggest that while women's autonomy has generally improved over time, the pattern of change was similar in both program and non-program states.

Figure 1 illustrates these dynamics more clearly. Between 2008 and 2013 before SURE-P's introduction, autonomy followed broadly parallel trajectories in both groups, supporting the parallel trends assumption required for the difference-in-differences design. After 2013, autonomy levels increased slightly in both treated and control states, but the gap between them was small. This pattern implies that the post-2013 improvement likely reflects broader socioeconomic progress, such as expanded education and increased women's participation in household decision-making, rather than a direct causal effect of the SURE-P intervention.

Table 2 reports the difference-in-differences regression estimates from both the baseline and controlled models. In the baseline model (column 1), autonomy shows modest gains over time but no statistically significant difference between treated and control states after the program's introduction. Once individual and regional characteristics are added (column 2), the results remain consistent. The interaction terms for the treatment and post-program periods treat#post2013 ($\beta = -0.0057$, $p = 0.57$) and treat#post2018 ($\beta = -0.0067$, $p = 0.48$) are small and statistically insignificant. This indicates that women's autonomy in SURE-P states did not change significantly relative to control states following program rollout.

However, the overall post-2018 effect is positive and significant ($\beta = 0.029$, $p < 0.01$), suggesting that women's autonomy improved nationwide between 2013 and 2018. These gains are thus more consistent with general socioeconomic developments such as rising female education, access to media, and greater participation in household decisions than with the direct influence of the SURE-P program.

The control variables behave as expected and align with prior literature. Age ($\beta = 0.005$, $p < 0.01$), education ($\beta = 0.067$, $p < 0.01$), and wealth ($\beta = 0.018$, $p < 0.01$) are all positively and significantly associated with autonomy, indicating that older, more educated, and wealthier women tend to have greater decision-making power. Urban residence is negatively related to autonomy ($\beta = -0.0018$, $p < 0.05$), which can be as a result of differences in household dynamics, composition and gender roles between rural and urban areas. Regional effects are pronounced: autonomy is lowest in the North East and North West and highest in the South South and South West, consistent with known cultural and educational disparities across these regions.

The full model explains about 28 percent of the variation in women's autonomy ($R^2 = 0.2755$), compared to less than 1 percent in the unadjusted model. This improvement highlights the importance of demographic and geographic heterogeneity in shaping women's empowerment outcomes in Nigeria. **Overall, while autonomy levels have gradually improved, there is no strong evidence that the SURE-P maternal CCT program directly enhanced women's autonomy. Instead, the observed progress appears to stem from broader structural and social transformations occurring nationwide.**

Conclusion and Policy Implications

This study examined the impact of Nigeria's SURE-P Maternal and Child Health conditional cash transfer program on women's decision-making autonomy using DHS data from 2008, 2013, and 2018. Although descriptive trends and regressions show that women's autonomy improved nationally over time, the difference-in-differences estimates reveal no significant treatment effect in SURE-P states relative to control states. These findings suggest that while the program successfully expanded maternal health-service utilization, it may not have translated into broader empowerment outcomes.

The results underscore that financial incentives alone are insufficient to transform gender relations. Sustained improvements in women's autonomy likely require complementary investments in education, access to information, and gender-inclusive community programs. Similar evidence from other contexts supports this conclusion: The authors (Baird et al, 2014) show that cash transfers improve welfare only when combined with schooling and social-norm interventions that empower girls.

In the Nigerian context, this implies that CCTs should be integrated with policies that directly enhance women's capacity to make informed health and economic decisions. Future research could further explore these dynamics to understand the social processes through which empowerment evolves.

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